CARBON MARKETS CCUS AND DIRECT AIR CAPTURE

CARBON-UNIT IDENTITY FORK → MARKET ARCHITECTURE MATRIX → CLAIM-TYPE LADDER → INTEGRITY GATE → CREDIT LIFECYCLE RAIL → ACCOUNTING-RISK FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◈ SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 CARBON-UNIT IDENTITY FORK

CARBON-UNIT IDENTITY FORK · PERMISSION UNIT INSIDE A COMPLIANCE CONSTRAINT · VERIFIED OUTCOME AGAINST AN APPROVED BASELINE · ALLOCATION OR AUCTION VERSUS METHODOLOGY AND ISSUANCE · SURRENDER OR RETIREMENT UNDER THE APPLICABLE RULE · TRADABILITY DOES NOT ERASE DIFFERENT UNIT ORIGINS

ALLOWANCE → permission unit inside a compliance constraint

CREDIT → verified outcome against an approved baseline → ORIGIN → allocation or auction versus methodology and issuance → USE → surrender or retirement under the applicable rule

SYSTEM / LEVEL / DEFINITION
- ALLOWANCE → permission unit inside a compliance constraint
- CREDIT → verified outcome against an approved baseline

STRUCTURE-FUNCTION MECHANISM
- ORIGIN → allocation or auction versus methodology and issuance
- USE → surrender or retirement under the applicable rule

SCALE / EVIDENCE LIMIT
- TRAP → tradability does not erase different unit origins
- ALLOWANCE → permission unit inside a compliance constraint

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → tradability does not erase different unit origins.

01

STAGE 01 MARKET ARCHITECTURE MATRIX

MARKET ARCHITECTURE MATRIX · CAP AND TRADE · ABSOLUTE COVERED-EMISSIONS CEILING · BASELINE AND CREDIT · PERFORMANCE AGAINST TARGET OR BASELINE · LEGAL OBLIGATION AND SURRENDER REQUIREMENT · NON-MANDATED PURCHASE OR CLAIM · WHAT CONSTRAINS EMISSIONS AND WHO MUST ACT

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CAP AND TRADE → absolute covered-emissions ceiling	COMPLIANCE → legal obligation and surrender requirement	QUESTION → what constrains emissions and who must act
BASELINE AND CREDIT → performance against target or baseline	VOLUNTARY → non-mandated purchase or claim	CAP AND TRADE → absolute covered-emissions ceiling

ECOLOGICAL UNIT / PARAMETER
- CAP AND TRADE → absolute covered-emissions ceiling
- BASELINE AND CREDIT → performance against target or baseline

DECISIVE DISTINCTION
- COMPLIANCE → legal obligation and surrender requirement
- VOLUNTARY → non-mandated purchase or claim

SCALE / STATUS / EXCEPTION
- QUESTION → what constrains emissions and who must act
- CAP AND TRADE → absolute covered-emissions ceiling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUESTION → what constrains emissions and who must act.

02

STAGE 02 CLAIM-TYPE LADDER

CLAIM-TYPE LADDER · COUNTERFACTUAL SOURCE DOES NOT ARISE · EXISTING SOURCE EMITS LESS · CARBON LEAVES AMBIENT ATMOSPHERE · DURATION AND REVERSAL RISK DETERMINE DURABILITY · NO MERGER · EACH SUPPORTS A DIFFERENT CLIMATE CLAIM

AVOIDANCE → counterfactual source does not arise

REDUCTION → existing source emits less → REMOVAL → carbon leaves ambient atmosphere → STORAGE → duration and reversal risk determine durability

SYSTEM / LEVEL / DEFINITION
- AVOIDANCE → counterfactual source does not arise
- REDUCTION → existing source emits less

STRUCTURE-FUNCTION MECHANISM
- REMOVAL → carbon leaves ambient atmosphere
- STORAGE → duration and reversal risk determine durability

SCALE / EVIDENCE LIMIT
- NO MERGER → each supports a different climate claim
- AVOIDANCE → counterfactual source does not arise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO MERGER → each supports a different climate claim.

03

STAGE 03 INTEGRITY GATE

INTEGRITY GATE · DEFINE THE COUNTERFACTUAL · TEST BUSINESS-AS-USUAL RISK · QUANTIFY BOUNDARY PERIOD AND OUTCOME · INDEPENDENT EVIDENCE CHECK · UNIT CREATED ONLY AFTER RULE-DEFINED GATES

BASELINE → define the counterfactual → ADDITIONALITY → test business-as-usual risk → MRV → quantify boundary period and outcome → VERIFICATION → independent evidence check → ISSUANCE → unit created only after rule-defined gates

SYSTEM INPUT / STARTING CONDITION
- BASELINE → define the counterfactual
- ADDITIONALITY → test business-as-usual risk

ECOLOGICAL PROCESS / TRANSFER
- MRV → quantify boundary period and outcome
- VERIFICATION → independent evidence check

OUTCOME / FEEDBACK / LIMIT
- ISSUANCE → unit created only after rule-defined gates
- BASELINE → define the counterfactual

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ISSUANCE → unit created only after rule-defined gates.

04

STAGE 04 CREDIT LIFECYCLE RAIL

CREDIT LIFECYCLE RAIL · DESIGN OR VALIDATION · METHOD AND PROJECT ACCEPTED · ACTIVITY AND OUTCOME RECORDED · EVIDENCE EXAMINED · ISSUANCE AND TRANSFER · TRADABLE UNIT ENTERS REGISTRY · UNIT REMOVED FROM CIRCULATION FOR A CLAIM

DESIGN OR VALIDATION → method and project accepted → MONITORING → activity and outcome recorded → VERIFICATION → evidence examined → ISSUANCE AND TRANSFER → tradable unit enters registry → RETIREMENT → unit removed from circulation for a claim

SYSTEM INPUT / STARTING CONDITION
- DESIGN OR VALIDATION → method and project accepted
- MONITORING → activity and outcome recorded

ECOLOGICAL PROCESS / TRANSFER
- VERIFICATION → evidence examined
- ISSUANCE AND TRANSFER → tradable unit enters registry

OUTCOME / FEEDBACK / LIMIT
- RETIREMENT → unit removed from circulation for a claim
- DESIGN OR VALIDATION → method and project accepted

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RETIREMENT → unit removed from circulation for a claim.

05

STAGE 05 ACCOUNTING-RISK FIREWALL

ACCOUNTING-RISK FIREWALL · DOUBLE ISSUANCE · MORE THAN ONE UNIT FOR ONE OUTCOME · DOUBLE USE · ONE UNIT USED MORE THAN ONCE · DOUBLE CLAIMING · SAME OUTCOME CLAIMED BY MULTIPLE ACTORS · CORRESPONDING ADJUSTMENT

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
DOUBLE ISSUANCE → more than one unit for one outcome	DOUBLE CLAIMING → same outcome claimed by multiple actors	RULE → adjustment is not issuance transfer or retirement
DOUBLE USE → one unit used more than once	CORRESPONDING ADJUSTMENT → international accounting treatment	DOUBLE ISSUANCE → more than one unit for one outcome

ECOLOGICAL UNIT / PARAMETER
- DOUBLE ISSUANCE → more than one unit for one outcome
- DOUBLE USE → one unit used more than once

DECISIVE DISTINCTION
- DOUBLE CLAIMING → same outcome claimed by multiple actors
- CORRESPONDING ADJUSTMENT → international accounting treatment

SCALE / STATUS / EXCEPTION
- RULE → adjustment is not issuance transfer or retirement
- DOUBLE ISSUANCE → more than one unit for one outcome

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjustment is not issuance transfer or retirement.

06

STAGE 06 INDIA CCTS STAGE LADDER

INDIA CCTS STAGE LADDER · LEGAL BASIS · ENABLING FRAMEWORK · SCHEME DESIGN · COMPLIANCE AND OFFSET PATHWAYS · TARGET OR METHODOLOGY · OBLIGATION AND CREDIT RULES · ISSUANCE AND TRADING

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways → TARGET OR METHODOLOGY → obligation and credit rules → ISSUANCE AND TRADING → operational transactions

SYSTEM / LEVEL / DEFINITION
- LEGAL BASIS → enabling framework
- SCHEME DESIGN → compliance and offset pathways

STRUCTURE-FUNCTION MECHANISM
- TARGET OR METHODOLOGY → obligation and credit rules
- ISSUANCE AND TRADING → operational transactions

SCALE / EVIDENCE LIMIT
- PRICE AND VOLUME → dated market evidence only
- LEGAL BASIS → enabling framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRICE AND VOLUME → dated market evidence only.

07

STAGE 07 CCUS CHAIN

CCUS CHAIN · CONCENTRATED CARBON DIOXIDE STREAM · CAPTURE AND CONDITIONING · SEPARATION AND PREPARATION · PIPELINE SHIP ROAD OR OTHER ROUTE · UTILISATION OR STORAGE · DESTINATION DETERMINES DURABILITY · MONITORING AND CLOSURE

SOURCE → concentrated carbon dioxide stream → CAPTURE AND CONDITIONING → separation and preparation → TRANSPORT → pipeline ship road or other route → UTILISATION OR STORAGE → destination determines durability → MONITORING AND CLOSURE → permanence and responsibility

SYSTEM INPUT / STARTING CONDITION
- SOURCE → concentrated carbon dioxide stream
- CAPTURE AND CONDITIONING → separation and preparation

ECOLOGICAL PROCESS / TRANSFER
- TRANSPORT → pipeline ship road or other route
- UTILISATION OR STORAGE → destination determines durability

OUTCOME / FEEDBACK / LIMIT
- MONITORING AND CLOSURE → permanence and responsibility
- SOURCE → concentrated carbon dioxide stream

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → pipeline ship road or other route.

08

STAGE 08 UTILISATION-STORAGE TEST

UTILISATION-STORAGE TEST · PRODUCT USE · HOW LONG IS CARBON RETAINED · PROCESS USE · DOES LATER RELEASE OCCUR · GEOLOGICAL STORAGE · CONTAINMENT AND MONITORING · REVERSAL RISK AND REMEDIATION

PRODUCT USE → how long is carbon retained

PROCESS USE → does later release occur → GEOLOGICAL STORAGE → containment and monitoring → LEAKAGE → reversal risk and remediation

SYSTEM / LEVEL / DEFINITION
- PRODUCT USE → how long is carbon retained
- PROCESS USE → does later release occur

STRUCTURE-FUNCTION MECHANISM
- GEOLOGICAL STORAGE → containment and monitoring
- LEAKAGE → reversal risk and remediation

SCALE / EVIDENCE LIMIT
- NET BENEFIT → full lifecycle accounting required
- PRODUCT USE → how long is carbon retained

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRODUCT USE → how long is carbon retained.

09

STAGE 09 CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON · POINT SOURCE CCUS · CARBON INTERCEPTED BEFORE RELEASE · CARBON SEPARATED FROM AMBIENT AIR · BIOGENIC CARBON PLUS CAPTURE AND STORAGE · REMOVAL TEST · ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE · LIFECYCLE TEST

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
POINT SOURCE CCUS → carbon intercepted before release	BECCS → biogenic carbon plus capture and storage	LIFECYCLE TEST → energy land transport and reversal
DAC → carbon separated from ambient air	REMOVAL TEST → atmospheric origin plus durable storage	POINT SOURCE CCUS → carbon intercepted before release

ECOLOGICAL UNIT / PARAMETER
- POINT SOURCE CCUS → carbon intercepted before release
- DAC → carbon separated from ambient air

DECISIVE DISTINCTION
- BECCS → biogenic carbon plus capture and storage
- REMOVAL TEST → atmospheric origin plus durable storage

SCALE / STATUS / EXCEPTION
- LIFECYCLE TEST → energy land transport and reversal
- POINT SOURCE CCUS → carbon intercepted before release

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LIFECYCLE TEST → energy land transport and reversal.

10

STAGE 10 NET-ZERO MITIGATION HIERARCHY

NET-ZERO MITIGATION HIERARCHY · PREVENT UNNECESSARY EMISSIONS · DECARBONISE EXISTING ACTIVITIES · ADDRESS HARD-TO-ABATE POINT SOURCES · COUNTERBALANCE GENUINELY RESIDUAL EMISSIONS · NET BALANCE NEEDS ONE BOUNDARY AND PERIOD

AVOID → prevent unnecessary emissions

REDUCE → decarbonise existing activities → CAPTURE → address hard-to-abate point sources → REMOVE → counterbalance genuinely residual emissions

SYSTEM / LEVEL / DEFINITION
- AVOID → prevent unnecessary emissions
- REDUCE → decarbonise existing activities

STRUCTURE-FUNCTION MECHANISM
- CAPTURE → address hard-to-abate point sources
- REMOVE → counterbalance genuinely residual emissions

SCALE / EVIDENCE LIMIT
- VERIFY → net balance needs one boundary and period
- AVOID → prevent unnecessary emissions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → net balance needs one boundary and period.

11

STAGE 11 CARBON INSTRUMENTS ANSWER SPINE

CARBON INSTRUMENTS ANSWER SPINE · UNIT MARKET AND CLAIM TYPE · BASELINE MRV ISSUANCE TRANSFER RETIREMENT · CAPTURE TRANSPORT UTILISATION OR STORAGE · PERMANENCE LEAKAGE LIFECYCLE AND DOUBLE COUNTING · CURRENT RULE PRICE VOLUME COST AND CAPACITY SOURCE

CLASSIFY → unit market and claim type → TRACE → baseline MRV issuance transfer retirement → FOLLOW → capture transport utilisation or storage → TEST → permanence leakage lifecycle and double counting → AUDIT → current rule price volume cost and capacity source

SYSTEM / LEVEL / DEFINITION
- CLASSIFY → unit market and claim type
- TRACE → baseline MRV issuance transfer retirement

STRUCTURE-FUNCTION MECHANISM
- FOLLOW → capture transport utilisation or storage
- TEST → permanence leakage lifecycle and double counting

SCALE / EVIDENCE LIMIT
- AUDIT → current rule price volume cost and capacity source
- CLASSIFY → unit market and claim type

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → current rule price volume cost and capacity source.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · DAC'S FUNDAMENTAL TECHNICAL CHALLENGE IS THERMODYNAMIC · ATMOSPHERIC CO_2 CONCENTRATION · POLICY IMPLICATION · BECAUSE DAC DELIVERS GENUINE ATMOSPHERIC REMOVAL (UNLIKE · GLOBAL CCUS AND DAC DEPLOYMENT CAPACITY REMAINS SMALL RELATIVE · THE INDIAN CARBON MARKET'S SPECIFIC SECTORAL COVERAGE, TARGET-SETTING METHODOLOGY · MRV-SYSTEM ROBUSTNESS FOR BOTH INDIA'S DOMESTIC CARBON MARKET AND

OPTIONAL ECOLOGICAL PROCESS DEPTH
- CAUTION: DAC's fundamental technical challenge is thermodynamic: atmospheric CO_2 concentration
- CAUTION: Policy implication: because DAC delivers genuine atmospheric removal (unlike
- CAUTION: Global CCUS and DAC deployment capacity remains small relative to the scale IPCC
- CAUTION: The Indian Carbon Market's specific sectoral coverage, target-setting methodology and

SCALE, PARAMETER OR STATUS LIMIT
- CAUTION: MRV-system robustness for both India's domestic carbon market and international
- India's carbon market has a two-track design: a compliance mechanism (building on PAT,
- Standard fossil-source CCUS is an emissions-avoidance technology; genuine atmospheric
- DAC's higher cost/energy-intensity relative to CCUS stems from the much lower CO_2

COMPARATIVE RESTORATION NUANCE
- Net-zero frameworks require genuine removal technologies (DAC/BECCS/nature-based sinks)
- NOT: All CCUS applications, including standard fossil-power-plant CCUS, deliver negative
- NOT: India's carbon market is a single-track, purely compliance-based system. - It has a
- NOT: DAC is currently deployed at a scale comparable to global CCUS capacity. - DAC remains

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels